Test performed

Board B's firmware was modified to always report its commanded angular velocity setpoint as the actual angular velocity. This test will show us how much of that "dead time" lives in the informational latency path, vs. how much of it could be improved by optimizing the PID controller and system dynamics.

The following topics were bagged:
ros2 bag record -o latency_bag1 /diffbot_base_controller/cmd_vel /diffbot_base_controller/odom /tf /dynamic_joint_states /tf_static /rosout /gpio_controller/gpio_states

The robot was given various angular commands using rqt's message publisher plugin. The sharp transitions should make it easier to see.

The robot was propped up during this test; the wheels were not contacting the ground. The URDF and diffbot base controller corrections were applied here; I assume that wouldn't affect the latency too much.

rtlatency

The "rtlatency" is reported in gpio_controller/gpio_states. The value spikes very high when the robot switches to manual mode; this is fine, you can just filter those out (if it's too high, ignore it). This shouldn't matter; the robot stayed in autonomous mode for the whole test.

In your report…

Just give me all the figures you gave in your previous latency report, minus the IMU data. Use diffbot_base_controller/cmd_vel as the command velocity reference. Call it "latency_report_informational.pdf".